Task 1: Enable CloudTrail and identify events

1. Open the AWS CloudTrail Console.
2. Open Event history. AWS provides a 90-day history of management events by default for each supported region without requiring you to create a trail.
3. In a test account, create an S3 bucket with a unique name.
4. Return to CloudTrail Event history and filter by Event source: s3.amazonaws.com.
5. Find the event associated with your bucket creation.
6. Perform another test action, such as creating a second bucket, and find the corresponding event.
7. Open each event to review its details.

Example event	What it records
CreateBucket	A bucket creation request and related event details.
PutBucketPolicy	A request to add or update a bucket policy, if you perform that action.
DeleteBucket	A bucket deletion request, if you perform that action.

Task 2: Find ConsoleLogin events from the last 24 hours

1. Open CloudTrail → Event history.
2. Set the time range to the last 24 hours.
3. Filter by Event name: ConsoleLogin.
4. Open each matching record.
5. Record the username, event time and whether the login succeeded or failed.

Use this table in your report:

Username	Timestamp (UTC)	Login result

Enter observed username	Enter observed timestamp	Success / Failure
Enter observed username	Enter observed timestamp	Success / Failure

Task 3: Importance of audit trails

Audit trails are records of actions performed on systems and resources. They help companies identify unauthorized changes, investigate security incidents and demonstrate accountability during compliance reviews.

Scenario 1: Unauthorized transaction-related access

If an attacker obtains an administrator's credentials and changes permissions on a storage resource, audit logs can help investigators identify the identity, time and API action involved. Without these logs, identifying the source of the change becomes more difficult.

Scenario 2: Exposure of customer data

If a cloud administrator accidentally makes a storage resource publicly accessible, audit logs can help establish who changed the configuration and when. Missing logs can delay incident response, weaken evidence and complicate the company's assessment of its legal reporting obligations.

Task 4: Export IAM changes to CSV

For AWS, use CloudTrail Event history to find management events related to IAM.
1. Filter the event source to iam.amazonaws.com.
2. Review relevant events, including CreateUser, DeleteUser, CreateAccessKey, AttachUserPolicy, PutUserPolicy and UpdateAssumeRolePolicy, as applicable.

3. Set the desired time range and inspect the results.
4. Download the available event history, then filter the exported data for IAM events in a spreadsheet application.
5. Save the filtered records as IAM_Changes.csv.

Task 5: Investigate suspicious activity in a Zomato-style application

1. Unusual login activity
- Suspicious sign: Unexpected console logins or repeated failed login attempts.
- Investigation: Filter for ConsoleLogin, inspect the identity, timestamp, source IP address and response details, and compare them with expected activity.
- Response: Escalate suspicious access, secure affected credentials and follow the incident response process.

2. Unexpected permission changes
- Suspicious sign: An identity receives unusually broad permissions or an IAM policy is modified unexpectedly.
- Investigation: Filter for iam.amazonaws.com, examine the event name, actor, request parameters and affected identity.
- Response: Verify the change with the authorized administrator and revoke unauthorized permissions.

3. Unexpected storage configuration changes
- Suspicious sign: An S3 bucket policy changes, public access controls are modified, or logging settings are disabled.
- Investigation: Filter for the S3 event source and inspect events such as PutBucketPolicy or PutBucketPublicAccessBlock.
- Response: Restore approved settings, preserve evidence and determine whether data may have been exposed.